GCFH-MDA: Graph Collaborative Filtering with Hypergraph Augmentation for miRNA-Drug Association Prediction

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Abstract. MicroRNAs (miRNAs) play a pivotal role in drug response regulation by modulating target gene expression. However, existing miRNA-drug association prediction methods remain challenges due to sparse interaction data and insufficient integration of biological priors. To address these, we propose GCFH-MDA, a graph collaborative filtering framework with hypergraph augmentation. The model formulates the task as collaborative filtering on a prior-guided bipartite interaction graph. To overcome sparsity, auxiliary hypergraphs are constructed from two complementary views: a regulatory view derived from miRNA-gene and drug-gene networks, and an embedding view based on sequence and structural features. Geodesic feature augmentation prioritizes biologically specific pathways, while contrastive learning aligns local and global representations within each view. Experiments on 8,720 known associations under five-fold cross-validation show that GCFH-MDA achieves 96.64% AUC and 96.04% AUPR, outperforming seven state-of-the-art baselines. This demonstrate the effectiveness of integrating biological priors and multimodal information for prioritizing miRNA–drug candidates.

Keywords: miRNA-drug association prediction, collaborative filtering, hypergraph representation learning, contrastive learning.

1 Introduction

Cancer remains a leading cause of death worldwide, imposing a severe global health burden [1]. Although chemotherapy is a cornerstone of cancer treatment, the genetic and epigenetic evolution of tumor cells often drives drug resistance, leading to treatment failure and poor prognosis [2]. This resistance arises from complex mechanisms such as abnormal drug uptake, altered drug targets, dysregulated cell cycles, and enhanced DNA repair [3].

MicroRNAs (miRNAs), small non-coding RNAs of approximately 22 nucleotides, are pivotal post-transcriptional regulators in tumorigenesis and drug response. Increasing evidence shows that miRNAs modulate drug resistance through drug transporter regulation, DNA damage repair, apoptosis, EMT, and metastasis-related pathways [4]. Dysregulated miRNA expression contributes to tumor progression, while miRNA mimics and antagomiRs show therapeutic potential [5]. miRNAs can either promote or reverse chemoresistance; for example, miR-20a enhances cisplatin resistance in gastric cancer by targeting CYLD [6], whereas miR-200c reverses multidrug resistance in breast cancer via MDR1 suppression [7].

Although experimental approaches are essential for characterizing miRNA-drug regulatory mechanisms, they remain costly and time-consuming for large-scale screening. This has spurred the development of miRNA-drug association resources for computational prioritization. Databases such as NoncoRNA [8], ncRNADrug [9], and ncDR [10] provide curated associations and support data-driven MDA prediction.

Advances in high-throughput profiling and graph representation learning have promoted the application of graph neural networks (GNNs) in biomolecular interaction prediction [11]. Existing MDA methods can be broadly categorized into three groups. The first category focuses on integrating sequence and structural data. For example, DLST-MDA [12] fuses multi-scale CNNs with GCNs, while NASMDR [13] uses GIN-based drug encoders and k-mer miRNA embeddings optimized via neural architecture search. The second emphasizes GNN architecture innovations, including DMGAT [14] with diffusion-enhanced hybrid GCN-GAT modeling, DARSFormer [15] with spectral graph transformation and dynamic attention, and GraphTGI [16] for heterogeneous regulatory networks via graph attention and bilinear decoders. The third explores advanced learning paradigms, represented by SSLMCD [17], unifying attribute and topological graphs through self-supervised contrastive learning and multi-view attention for multi-task ncRNA association prediction.

Despite this progress, existing methods still face several limitations. First, known MDA associations are sparse, producing incomplete and noisy bipartite graphs. Second, miRNA-drug relationships are mediated by complex gene regulatory mechanisms that require multi-source biological knowledge. Third, although different biological modalities capture complementary macro- and micro-level information, many models still lack effective fusion and cross-modal alignment [18].

To address these challenges, we propose GCFH-MDA, a graph collaborative filtering framework integrating heterogeneous biological data with contrastive learning for intra-view alignment. Specifically, our contributions are:

- We formulate miRNA-drug association prediction as a prior-guided graph collaborative filtering problem, enabling modeling of both direct and high-order interaction patterns within a unified framework.
- We propose a dual-view hypergraph augmentation strategy that integrates regulatory and embedding information at both local and global scales, providing complementary structural and attribute-level representations.

- We introduce a geodesic feature refinement mechanism with hub-aware penalties to emphasize biologically meaningful paths and alleviate bias introduced by high-degree nodes.

2 Material and Methods

This section outlines the data preparation and GCFH-MDA architecture (Fig. 1).

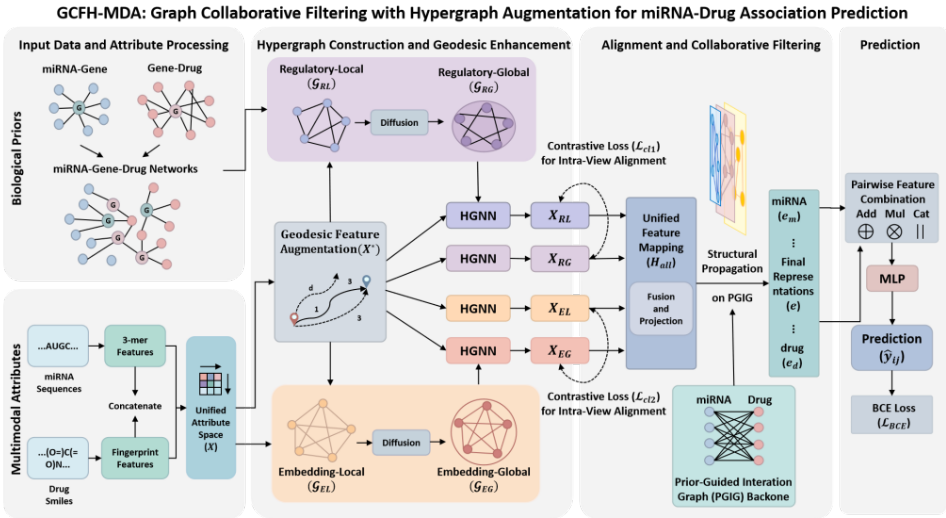

Fig. 1 Overview of GCFH-MDA.

2.1 Human miRNA-Drug Association Dataset

The benchmark was curated by Sheng et al. [18] from ncRNADrug [9], excluding deprecated miRNAs in miRBase v22 [19] and drugs with ambiguous DrugBank v5.1.12 annotations [20]. Mature miRNA sequences and drug SMILES were retrieved. The final dataset includes 8,720 validated associations between 1,578 miRNAs and 156 drugs. Additionally, miRNA-target and drug-target gene interactions were obtained from Sheng et al., encompassing 14,456 high-confidence interactions validated in miRTarBase [21] and the Comparative Toxicogenomics Database (CTD) [22].

2.2 Multimodal Node Attributes

MiRNA and drug features are encoded by mapping sequence and structural information into a shared d-dimensional space. MiRNA sequences are represented as normalized 3-mer frequency vectors [23], with cosine similarity forming the miRNA similarity matrix $S_m \in \mathbb{R}^{N_m \times N_m}$, where N_m is the number of miRNAs. Drugs are encoded with Morgan fingerprints [24] from SMILES, and Tanimoto similarity constructs the drug structural similarity matrix $S_d \in \mathbb{R}^{N_d \times N_d}$, where N_d is the number of drugs. Rows of S_m and S_d are used as initial node features and learnable transformations W_m and W_d project them into a shared latent space:

$$x_m^i = s_m^i \cdot W_m, \quad x_d^j = s_d^j \cdot W_d. \tag{1}$$

where s_m^i and s_d^j denote the i-th and j-th rows of S_m and S_d respectively. These resulting embeddings are concatenated into a feature matrix $X \in \mathbb{R}^{(N_m+N_d)\times d}$.

2.3 Heterogeneous Graph: Regulatory and Embedding Views

As shown in Fig. 1, four hypergraphs are constructed from regulatory and embedding views, including local ($\mathcal{G}_{RL}$, $\mathcal{G}_{EL}$) and global ($\mathcal{G}_{RG}$, $\mathcal{G}_{EG}$) hypergraphs.

Local Hypergraph. To capture high-order neighborhood relations, two local hypergraphs are constructed [25]. In the regulatory view, $\mathcal{G}_{RL} = (V, E_{RL})$ is built from top-k_g gene-centered associations, where nodes sharing the same gene form a hyperedge, yielding incidence matrix H_{RL}. In the embedding view, $\mathcal{G}_{EL} = (V, E_{EL})$ is constructed by connecting each node with its top-k_n nearest neighbors in embedding space, producing incidence matrix H_{EL}.

Global Hypergraph. To model long-range dependencies, $\mathcal{G}_{RG}$ and $\mathcal{G}_{EG}$ are generated from $\mathcal{G}_{RL}$ and $\mathcal{G}_{EL}$ via the same symmetric residual diffusion mechanism. For each view, the initial structural association matrix is $S^{(0)} = H_l^{(v)} D_e^{-1} (H_l^{(v)})^\top$, with symmetric diffusion iteratively defined as:

$$S^{(\mathrm{k})} = \alpha S^{(0)} + (1-\alpha)\,\Theta S^{(\mathrm{k}-1)} \tag{2}$$

where $\Theta = D_v^{-1/2} S^{(0)} D_v^{-1/2}$ and $\alpha \in (0,1)$ controls the contribution of the original structure [26]. After K-step diffusion, each node connects to its top-k_n most similar nodes to form the global incidence matrix H_{RG} and H_{EG}. Thus, local hypergraphs preserve fine-grained high-order associations, while global hypergraphs capture complementary long-range dependencies.

2.4 Feature Enhancement and Hypergraph Encoding

As shown in the feature enhancement module of Fig. 1, each hypergraph undergoes geodesic feature augmentation followed by hypergraph neural network encoding.

Geodesic Feature Augmentation. Following the generalized geodesic distance framework [27], we adapt it to hypergraph-based biological networks by introducing a hub-aware penalty to suppress nonspecific high-degree nodes and emphasize biologically meaningful paths. For each hypergraph $\mathcal{G}_v \in \{\mathcal{G}_{RL}, \mathcal{G}_{RG}, \mathcal{G}_{EL}, \mathcal{G}_{EG}\}$, geodesic features are obtained via a minimum-cost propagation process. We define a degree-based potential $\rho(i)$ to penalize paths through hubs:

$$\rho(i) = \left(\frac{d(i)}{\mathbb{E}[d]}\right)^{\gamma} \tag{3}$$

where $d(i)$ is the hyper-degree of v_i, $\mathbb{E}[d] = \frac{1}{|V|}\sum_i d\,(i)$ is the average hyper-degree, and γ is a specificity parameter. With scaling factor $\beta > 0$, the generalized geodesic distance is updated as:

$$f^{(t+1)}(i) = \begin{cases} \min\limits_{j \in \mathcal{N}_H(i)} \left(f^{(t)}(j)\right) + \beta\rho(i), & |\mathcal{N}_H(i)| > 0 \\ C_{max}, & |\mathcal{N}_H(i)| = 0 \end{cases} \tag{4}$$

where $\mathcal{N}_H(\mathrm{i})$ contains nodes sharing at least one hyperedge with v_i, and $C_{\max}$ applies to isolated nodes. Boundary nodes $\mathcal{B}$ are defined as seed nodes incident to the core hyperedges used to initialize geodesic propagation. The initial state is:

$$f^{(0)}(i) = \begin{cases} 0, & i \in \mathcal{B}, \\ +\infty, & otherwise \end{cases} \tag{5}$$

After convergence, $f_v(\mathrm{i})$ represents the geodesic distance from $\mathcal{B}$ to v_i. For each view, normalized $f_v(\mathrm{i})$ is concatenated with node feature x_i to form the augmented matrix X_v^* with rows $x_{v,i}^* = [x_i \,||\, \tilde{f}_v(i)]$.

Hypergraph Representation Learning. We apply Hypergraph Neural Networks [28] to process the augmented features. The convolution at the $(l+1)$-th layer is:

$$X_v^{(l+1)} = \phi\left(D_v^{-1/2} H_v D_e^{-1} H_v^{\top} D_v^{-1/2} X_v^{(l)} W_v^{(l)}\right) \tag{6}$$

where H_v is the incidence matrix of $\mathcal{G}_v$, D_v and D_e are node and hyperedge degree matrices, and $W_v^{(l)}$ is a learnable parameter. The four hypergraph branches yield:

$$\begin{aligned} X_{RL} &= \mathrm{HGNN}(X_{RL}^*, H_{RL}), X_{RG} = \mathrm{HGNN}(X_{RG}^*, H_{RG}), \\ X_{EL} &= \mathrm{HGNN}(X_{EL}^*, H_{EL}), X_{EG} = \mathrm{HGNN}(X_{EG}^*, H_{EG}) \end{aligned} . \tag{7}$$

2.5 Intra-View Alignment and Graph Collaborative Filtering

As shown in Fig. 1, regulatory and embedding views are aligned and refined the through the Prior-Guided Interaction Graph (PGIG) backbone.

Intra-View Contrastive Learning. Contrastive learning is applied between local and global representations within both regulatory and embedding views. For each view $v \in \{RL, RG, EL, EG\}$, node embeddings are projected into a latent space via an MLP: $Z_v = \mathrm{MLP}(X_v)$. For paired views (v, u), each node i has a positive pair (z_i^v, z_i^u) and negatives (z_i^v, z_k^u) for $k \neq i$. The symmetric InfoNCE loss is:

$$\mathcal{L}_{v,u} = -\frac{1}{2N}\sum_{i=1}^{N}\sum_{(a,b)\in\{(v,u),(u,v)\}} log \frac{exp(sim(z_i^a, z_i^b)/\tau)}{\sum_{k=1}^{N} exp(sim(z_i^a, z_k^b)/\tau)} . \quad (8)$$

where N is the number of nodes, $sim(\cdot)$ is cosine similarity function, and τ is a temperature hyperparameter. We define two contrastive losses: $\mathcal{L}_{cl1}$ for the regulatory perspective and $\mathcal{L}_{cl2}$ for the embedding perspective.

Inter-Space Fusion and PGIG Backbone Refinement. To integrate heterogeneous signals local and global representations within each perspective are first concatenated: $H_R = [X_{RL} \,||\, X_{RG}]$, $H_E = [X_{EL} \,||\, X_{EG}]$. We then apply MLP transformations to project them into a shared latent space: $\widetilde{H}_R = \mathrm{MLP_g}(H_R)$, $\widetilde{H}_E = \mathrm{MLP_E}(H_E)$, and the unified feature matrix is $H_{all} = [\widetilde{H}_R \,||\, \widetilde{H}_E]$.

Known miRNA-drug associations are encoded as a bipartite Prior-Guided Interaction Graph (PGIG). Let $Y \in \{0,1\}^{N_m \times N_d}$ denote the observed miRNA-drug association matrix. The PGIG adjacency matrix is defined by the block matrix: $A = \begin{bmatrix} 0 & Y \\ Y^{\top} & 0 \end{bmatrix} \in \mathbb{R}^{(N_m+N_d)\times(N_m+N_d)}$. To refine node representations, we adopt a collaborative filtering backbone with LightGCN propagation [29]. Specifically, we apply symmetric degree normalization: $\tilde{A} = D^{-1/2}AD^{-1/2}$, where D is the degree matrix of A. Starting from $E^{(0)} = H_{all}$, propagation is defined as:

$$E^{(l+1)} = \tilde{A}E^{(l)} \quad (9)$$

Following LightGCN, we obtain the final representation by averaging embeddings from all layers to capture high-order collaborative signals:

$$E_{final} = \frac{1}{L+1}\sum_{l=0}^{L} E^{(l)} \quad (10)$$

2.6 Prediction of Potential miRNA-Drug Associations

To capture complex interaction patterns between a given miRNA-drug pair (m_i, d_j), we extract their final representations e_{m_i} and e_{d_j} and form a pairwise feature vector $p = [e_{m_i} \,||\, e_{d_j}]$. The association probability $\hat{y}_{ij}$ is estimated as $\hat{y}_{ij} = \sigma(\mathrm{MLP}(p))$ via a layer MLP. The final loss function is:

$$\mathcal{L}_{total} = \mathcal{L}_{BCE} + \lambda_1\mathcal{L}_{cl1} + \lambda_2\mathcal{L}_{cl2} \quad (11)$$

where $\mathcal{L}_{BCE}$ is the binary cross-entropy loss between predicted scores $\hat{y}_{ij}$ and labels. Parameters λ_1 and λ_2 weight the contrastive losses.

3 Results

Experiments were conducted under five-fold cross-validation to evaluate GCFH-MDA, including baseline comparisons, ablation, hyperparameter sensitivity, cold-start evaluation, and case studies. The model was implemented in Python 3.8 and PyTorch v2.1.0 with CUDA on an NVIDIA RTX 4090 GPU. Adam was used with learning rate 0.001, batch size 128, and training for five epochs.

3.1 Evaluation Metrics

Verified miRNA-drug associations were treated as positives, with an equal number of negatives sampled from unknown pairs. Performance was assessed using Precision, Accuracy, Recall, F1-score, AUC, and AUPR. In each fold, the PGIG was constructed from training data only. As shown in Table 1, GCFH-MDA achieved an average AUC of 96.64% ± 0.26 and AUPR of 96.04% ± 0.34.

Table 1. Performance of GCFH-MDA under five-fold cross-validation.

Fold	AUC(%)	AUPR(%)	Precision(%)	Accuracy(%)	Recall(%)	F1-score(%)
1	96.37	95.76	87.04	90.39	93.77	90.67
2	96.57	95.75	86.09	91.25	96.35	91.72
3	97.13	96.66	90.59	91.42	92.29	91.35
4	96.61	95.86	89.05	91.33	93.67	91.44
5	95.51	96.12	88.79	91.56	94.19	91.98
Mean	**96.64±0.26**	**96.04±0.34**	**88.95±1.71**	**91.19±0.42**	**94.05±1.42**	**91.43±0.47**

3.2 Comparison with State-of-the-Art Methods

GCFH-MDA was compared with seven representative methods (DLST-MDA [12], NASMDR [13], DMGAT [14], DARSFormer [15], GraphTGI [16], SSLMCD [17], and MGCNA [18]) under five-fold cross-validation. All baselines were re-trained using official codes. As shown in Fig. 2, GCFH-MDA achieves the best performance on most metrics, improving AUC and AUPR by 0.70% and 0.85% over SSLMCD. with statistically significant gains ($p < 0.05$). These improvements benefit from integrating biological priors, multimodal information, and geodesic augmentation.

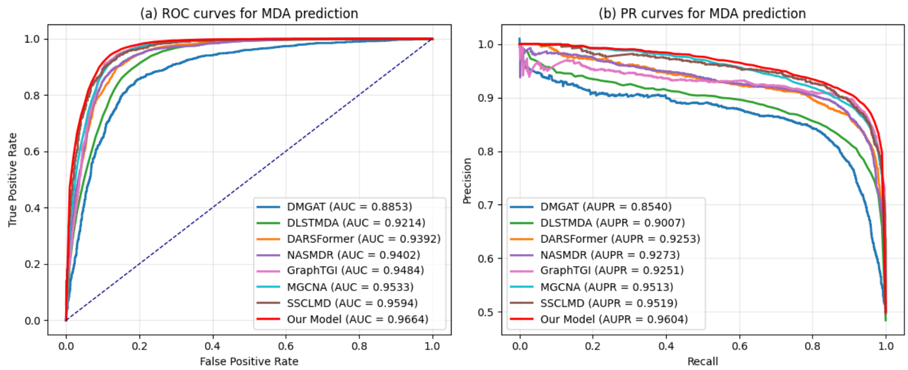

Fig. 2 Performance comparison of GCFH-MDA with baseline methods.

3.3 Ablation Study

Ablation experiments were conducted to assess the contribution of each component. As shown in Fig. 3, the full model is compared with several degraded variants:

- **w/o Reg.** Removing the regulatory hypergraph reduces AUC to 96.04% (↓0.60%), showing the contribution of gene-mediated priors.
- **w/o Emb.** Eliminating the embedding hypergraph lowers AUC to 95.47% (↓1.07%), indicating complementary patterns captured by multimodal attributes.
- **w/o Global.** Removing global structures and intra-view alignment decreases AUC to 96.03\% (↓0.61\%), reflecting the benefit of higher-order relations.
- **w/o Geo.** Without geodesic feature augmentation, the AUC drops to 96.11% (↓0.53%), showing its effectiveness in alleviating structural sparsity.
- **w/o Prop.** Removing the graph CF propagation causes a significant drop (AUC: 94.51%, ↓2.13%), highlighting the central role of collaborative filtering.

Overall, all components contribute positively, while the significant drop without propagation confirms graph collaborative filtering as the core module, with other components providing complementary gains.

3.4 Hyperparameter Sensitivity Analysis

We evaluated GCFH-MDA under key hyperparameters, including the number of core genes k_g, hyperedge scale k, diffusion coefficient α, geodesic weight β, contrastive temperature τ, propagation depth L, and contrastive loss weights λ_1, λ_2.

As shown in Fig. 4 and Fig. 5, the model maintains stable AUC and AUPR across a wide range of settings. Optimal performance is achieved at $k_g = 500$, $k = 10$, $\alpha = 0.7$, $\beta = 1.0$, $\tau = 0.5$, $L = 3$, and moderate contrastive weights ($\lambda_1 = 0.1$, $\lambda_2 = 0.5$). Results indicate embedding-based contrastive learning provides stronger regularization than regulatory view, while structural propagation and geodesic augmentation improve representations without over-smoothing.

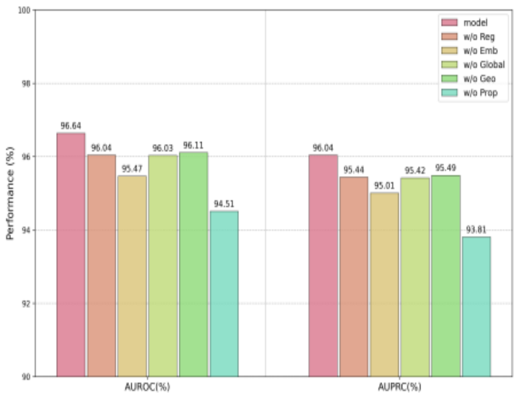

Fig. 3 Ablation study of GCFH-MDA.

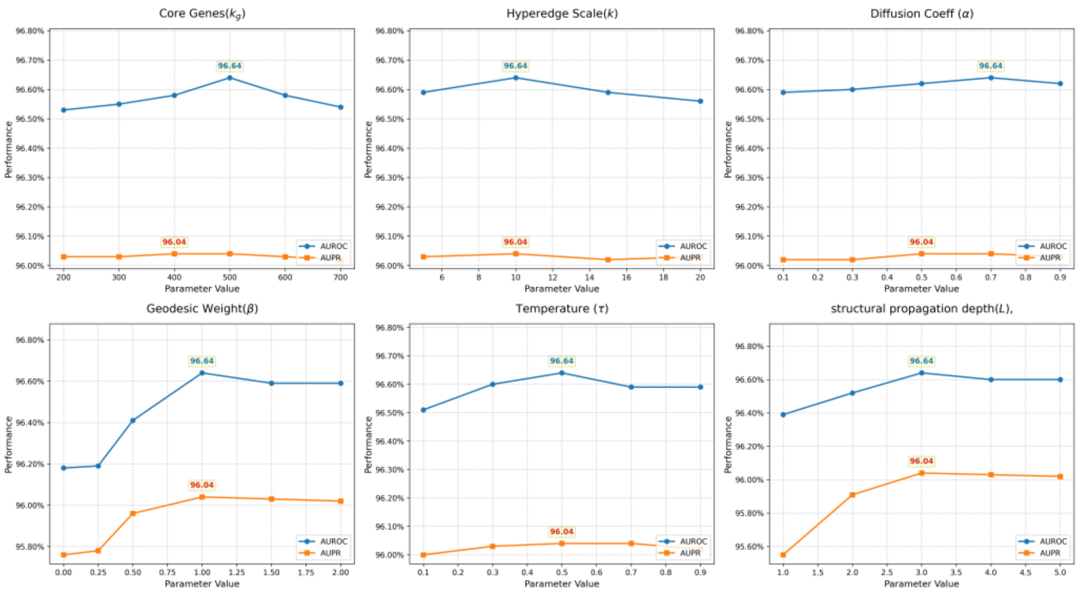

Fig. 4 Hyperparameter sensitivity analysis of GCFH-MDA.

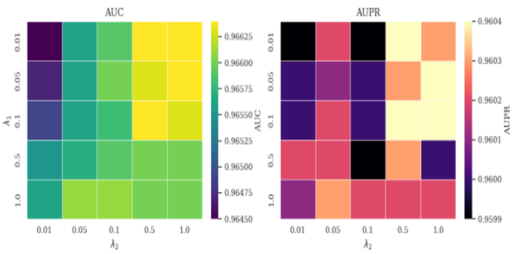

Fig. 5 Loss weight sensitivity analysis of GCFH-MDA

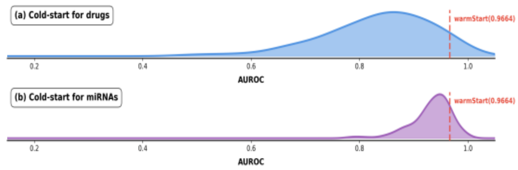

Fig. 6 Cold-start performance of GCFH-MDA

3.5 Performance in Cold-Start Scenarios

We further evaluate GCFH-MDA under cold-start settings: (i) drug cold-start and (ii) miRNA cold-start, where all associations of a held-out node are excluded for testing. The model is trained on remaining associations and evaluated on withheld interactions. As shown in Fig. 6, drug cold-start yields a broader AUC distribution (83.34%±9.83), while miRNA cold-start shows a narrower distribution (93.73%±3.98). This results support the framework's effectiveness in cold-start prediction, particularly for emerging miRNAs and novel drugs.

3.6 Case studies

To evaluate GCFH-MDA for prioritizing novel miRNA-drug associations, we conducted case studies on Fluorouracil, Cisplatin, and Doxorubicin. Using all known associations for training, the top-15 predicted miRNAs per drug were ranked by probability. As shown in Fig. 7, 7 candidates for 5-FU and cisplatin, and 6 for doxorubicin, are supported by literature [30-32]. These results demonstrate GCFH-MDA effectively identifies biologically relevant associations and may support drug resistance analysis and personalized treatment.

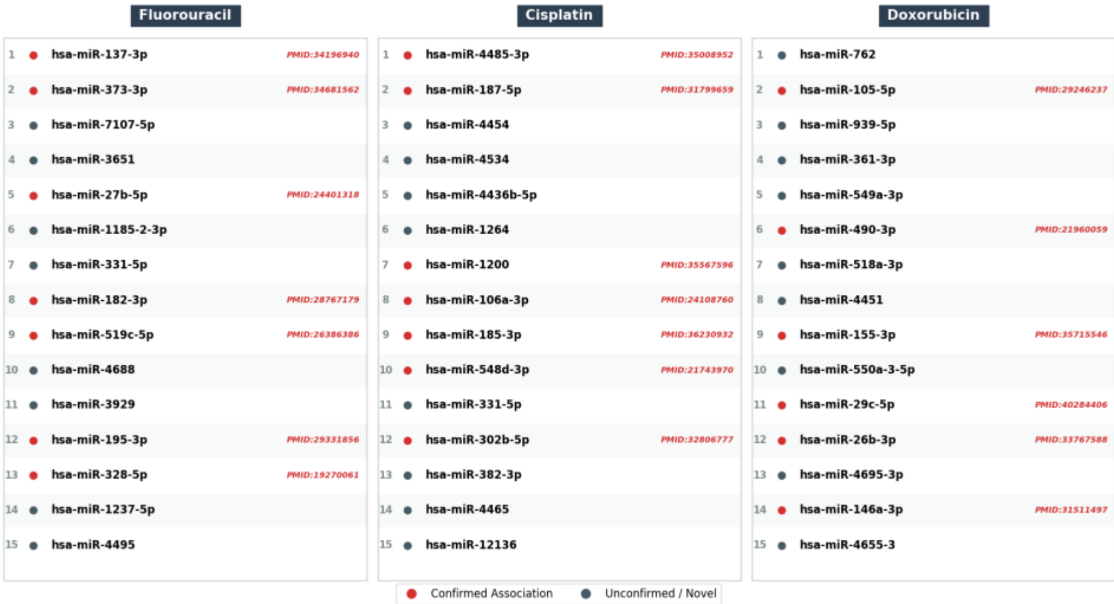

Fig. 7 Top 15 predicted miRNA-drug associations (not in the benchmark database) for Fluorouracil, Cisplatin, and Doxorubicin.

4 Conclusion

Drug response variability influenced by miRNAs is critical for therapeutic outcomes, making accurate identification of miRNA-drug associations essential for studying drug resistance and precision medicine. We proposed GCFH-MDA, a graph collaborative filtering framework with hypergraph augmentation that integrates biological priors and multimodal attributes. Contrastive learning and geodesic augmentation enhance representation learning, achieving state-of-the-art AUC and AUPR on 8,720 associations, further supported by case studies on fluorouracil, cisplatin, and doxorubicin. Future work will incorporate multi-omics and temporal data and experimentally validate mechanisms for personalized cancer therapy.

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